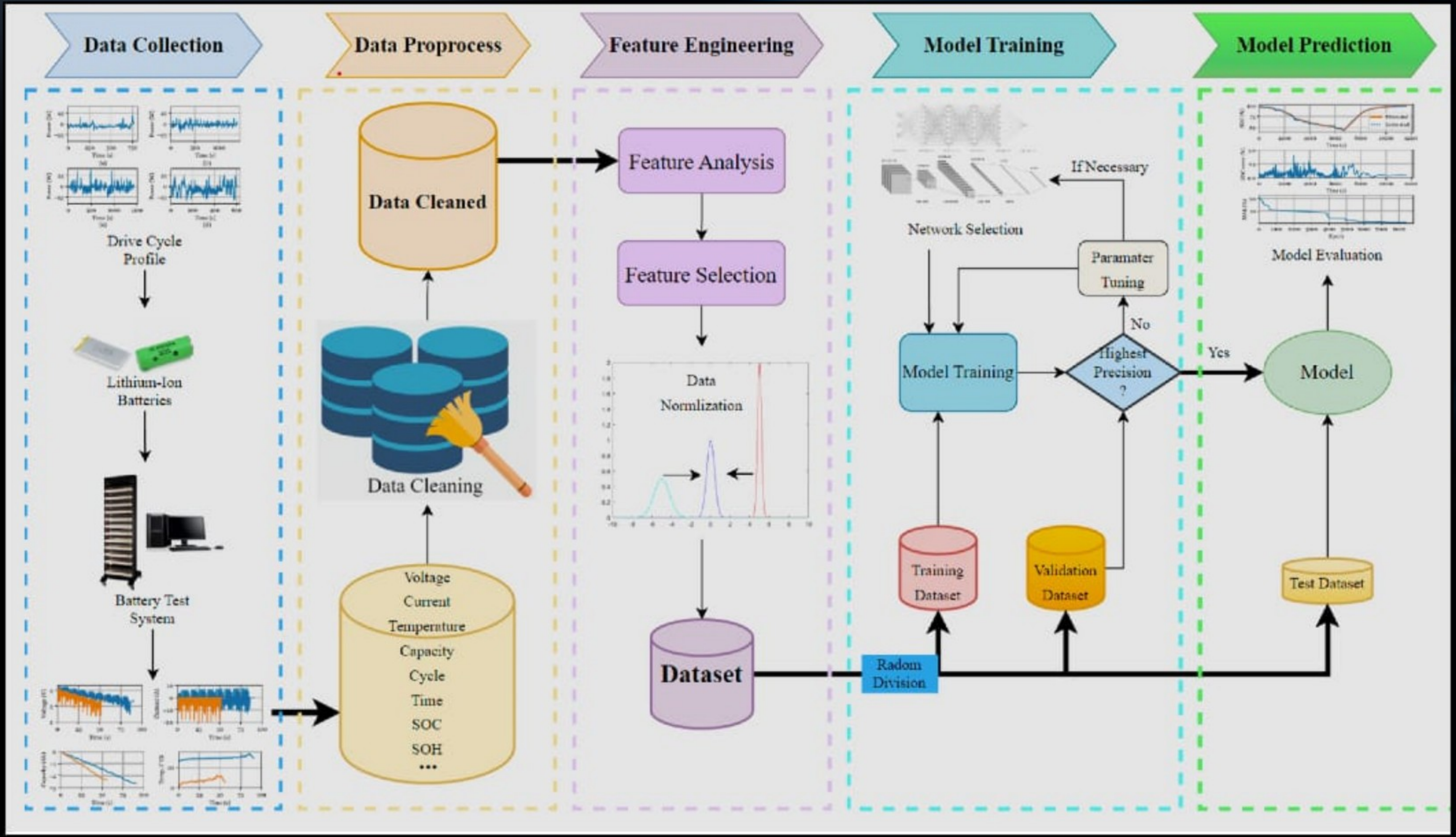
Battery Data Analysis & Feature Engineering Insights



NAME:- SREEJA SANYAL

WEEK :-2

DAY:-14

DATE:- 09.09.2026

EDA FINDINGS :

What did the data tell us?

1. Voltage, Current & Temperature → Key battery measurement variables
2. Power → derived electrical feature
3. Time → Charging progression
4. SOC → Primary prediction target
5. EDA revealed nonlinear relationships among several battery variables.
6. Data quality & leakage checks are important before model training.

```
BASIC EDA FINDINGS

Basic statistics:
          Voltage     Current  Temperature
count  845.000000  845.000000   845.000000
mean     3.798611    0.777509    24.159459
std      0.489803    1.221568     0.467085
min      2.908080    0.000000    23.345190
25%      3.168840    0.000000    23.870990
50%      4.185630    0.000000    23.976150
75%      4.193550    1.443070    24.501940
max      4.199950    3.001080    27.446380

Minimum values:
Voltage          2.90808
Current              0.0
Temperature     23.34519
Capacity               0
WhAccu                 0
dtype: object

Maximum values:
Voltage          4.19995
Current          3.00108
Temperature     27.44638
Capacity            [Ah]
WhAccu              [Wh]
dtype: object

Number of charging records:
358

Battery status values:
Status
PAU    486
CHA    358
STO      1
Name: count, dtype: int64
```

EDA was used to understand the distribution, trends, and relationships present in the battery data.

FEATURE ENGINEERING

Features Considered	FeaturePurpose
Voltage	Electrical measurement
Current	Charging measurement
Temperature	Thermal measurement
Power	Derived electrical feature(voltage *current)
SOC	Target/battery state variable

```
soc_raw_features = ["Voltage","Current","Temperature","Elapsed Time","Cycle"]
time_raw_features = ["Voltage","Current","Temperature", "Elapsed Time","Cycle","Capacity"]
if "WhAccu" in soc_train.columns:
    soc_raw_features.append("WhAccu")
if "WhAccu" in time_train.columns:
    time_raw_features.append("WhAccu")
soc_raw_features = [
    col for col in soc_raw_features
    if col in soc_train.columns]
time_raw_features = [
    col for col in time_raw_features
    if col in time_train.columns]
print("SOC raw features:")
print(soc_raw_features)
print("\nRemaining-time raw features:")
print(time_raw_features)
```

```
SOC raw features:
['Voltage', 'Current', 'Temperature', 'Elapsed Time', 'Cycle', 'WhAccu']

Remaining-time raw features:
['Voltage', 'Current', 'Temperature', 'Elapsed Time', 'Cycle', 'Capacity', 'WhAccu']
```

Relevant electrical and time-based features were derived to support SOC and charging-time prediction.

DATA PREPARATION FOR MODELING

Component	Selected Variable
Input Features (X)	Voltage, Current, Temperature, Power
Target (y)	Remaining Charging Time
Data Split	Training Set + Test Set
Purpose	Prepare the dataset for the upcoming regression stag

```
TARGET VARIABLES:
1. SOC
2. Remaining Charging Time

SOC INPUT VARIABLES:
- Voltage
- Current
- Temperature
- Elapsed Time
- Cycle
- WhAccu

REMAINING-TIME INPUT VARIABLES:
- Voltage
- Current
- Temperature
- Elapsed Time
- Cycle
- Capacity
- WhAccu

SOC DATA SPLIT:
Training   : 9926
Validation : 7376
Testing    : 2250

REMAINING-TIME DATA SPLIT:
Training   : 41015
Validation : 1349
Testing    : 2286

PREPROCESSING:
- Missing values: median imputation
- Scaling: StandardScaler
- Fitted only on training data
- Fixed random seed: 42
- Group-based leakage check: PASSED

OUTPUT DIRECTORY:
/content/drive/MyDrive/BATTERY_SOC_PROJECT/day11_outputs
```

The target variables and selected input features were summarized before model development.

3.

Prediction Setup

Input Features (X)
(Voltage + Current + Temperature + Power)
↓
Regression Model
↓
Target (y)
Remaining Charging Time

- The selected charging-related variables are used as input features.
- Remaining Charging Time is defined as the prediction target.
- This creates the basic input–target structure required for regression.
- The prepared setup will be used in the upcoming modeling stage.

```python
# X AND y FOR SOC
X_soc_train = soc_train[soc_input_variables].copy()
X_soc_validation = soc_validation[soc_input_variables].copy()
X_soc_test = soc_test[soc_input_variables].copy()
y_soc_train = soc_train[["SOC"]].copy()
y_soc_validation = soc_validation[["SOC"]].copy()
y_soc_test = soc_test[["SOC"]].copy()
```

```python
# X AND y FOR REMAINING TIME

X_time_train = time_train[time_input_variables].copy()
X_time_validation = time_validation[time_input_variables].copy()
X_time_test = time_test[time_input_variables].copy()
y_time_train = time_train[["Remaining Charging Time"]].copy()
y_time_validation = time_validation[["Remaining Charging Time"]].copy()
y_time_test = time_test[["Remaining Charging Time"]].copy()
```

Separate input features (X) and target variables (y) were defined for SOC estimation and Remaining Charging Time prediction.

Train–Test Split

Why was the dataset split?

- Training set → Used for learning patterns from the available data.
- Test set → Used to evaluate the model on data not used during training.
- Separating the two sets helps prevent evaluating the model on the same observations it learned from.
- The test set provides a more realistic indication of how the model may perform on unseen data.
- This separation is an important step before developing the regression models.

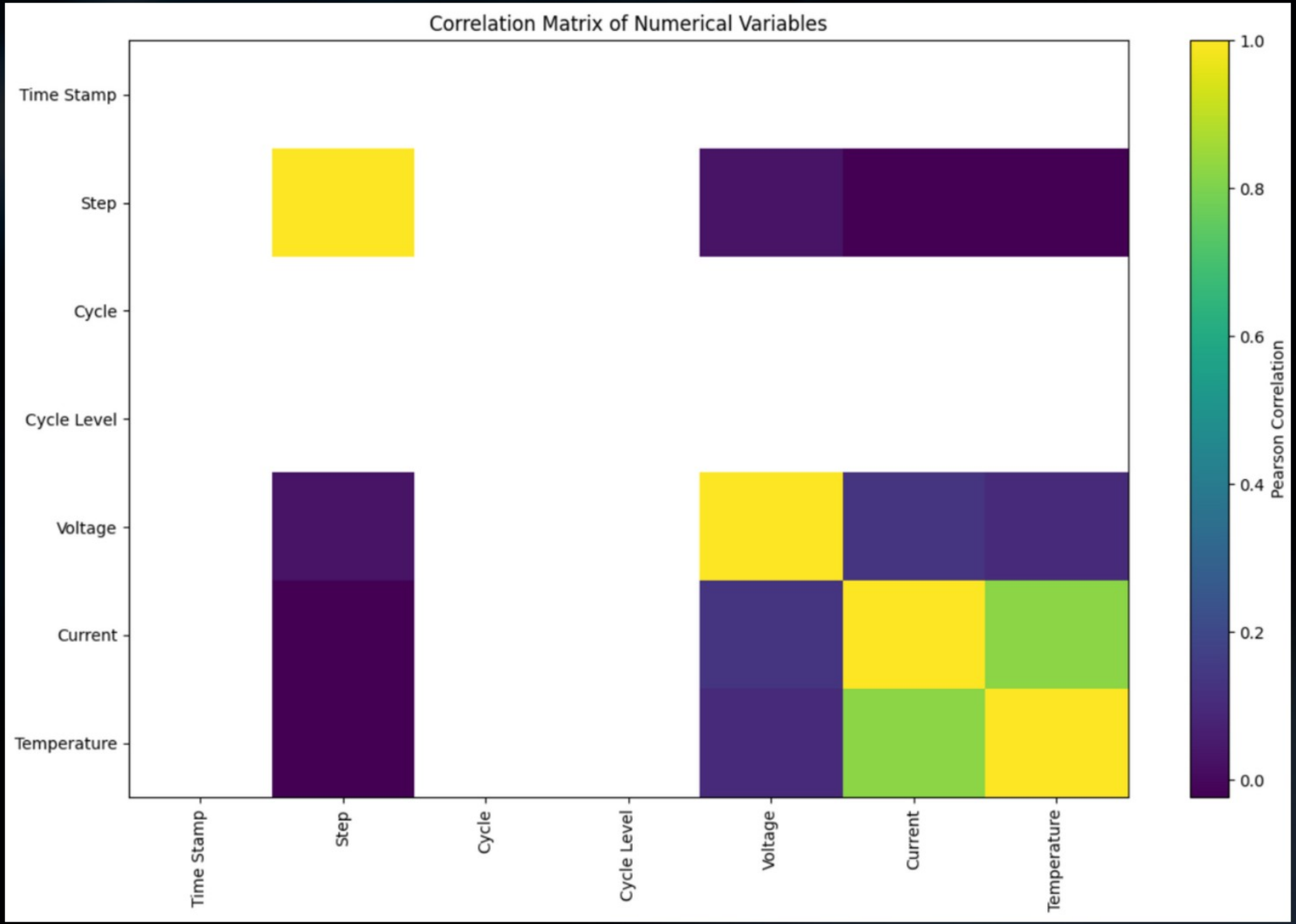
```
TRAIN / VALIDATION

SOC split sizes:
Train: (9926, 9)
Validation: (7376, 9)
Test: (2250, 9)

Remaining-time split sizes:
Train: (41015, 10)
Validation: (1349, 10)
Test: (2286, 10)

Total SOC rows: 19552
Total remaining-time rows: 44650
```

The dataset was divided into training, validation, and testing sets for reliable model evaluation.”

Correlation Analysis

What was checked?

- Examined relationships between the numerical variables in the dataset.
- Checked how the selected input features are related to each other.
- Examined the relationship between the predictors and the prediction target.
- Correlation helps identify the strength and direction of linear relationships.
- The analysis provides an initial understanding of variable dependencies before regression modeling.
- Correlation was treated as an exploratory analysis rather than proof of causation.



Correlation analysis was performed to examine relationships between the selected features and target variables.

Data Leakage Check

Why was leakage checked?

- Checked whether information related to the target could unintentionally appear among the input variables.
- Data leakage can make a model appear more accurate than it actually is.
- A proper modeling setup should ensure that the model receives only appropriate predictor information.
- The check helps maintain a fair separation between input features and the prediction target.
- This improves the reliability of the evaluation in the upcoming modeling stage.

```
BASIC DATA LEAKAGE REVIEW

SOC target:
SOC
SOC input features:
['Voltage', 'Current', 'Temperature', 'Elapsed Time', 'Cycle', 'WhAccu']

Remaining Charging Time target:
Remaining Charging Time
Remaining-time input features:
['Voltage', 'Current', 'Temperature', 'Elapsed Time', 'Cycle', 'Capacity', 'WhAccu']

✓ SOC is not included as an SOC input feature.
✓ Remaining Charging Time is not an input feature.

Source_File overlap checks:
SOC Train vs SOC Validation: 0 overlapping source files
SOC Train vs SOC Test: 0 overlapping source files
SOC Validation vs SOC Test: 0 overlapping source files
Time Train vs Time Validation: 0 overlapping source files
Time Train vs Time Test: 0 overlapping source files
Time Validation vs Time Test: 0 overlapping source files
```

The selected features were checked to ensure that target information was not unintentionally included.

Multicollinearity Check

Correlation-Based Check

- Examined the correlation between selected input features.
- Separate correlation matrices were generated for the SOC feature set and Remaining Charging Time feature set.
- Power was included along with the respective raw features.
- The correlation values were rounded for easier interpretation.
- This check helps identify features that may have strong relationships with each other.
- It provides an initial indication of possible multicollinearity before regression modeling.

SOC Features Correlation:

	Voltage	Current	Temperature	Elapsed Time	Cycle	WhAccu	Power
Voltage	1.00	0.00	0.22	0.04	NaN	0.81	0.04
Current	0.00	1.00	0.33	-0.52	NaN	0.35	1.00
Temperature	0.22	0.33	1.00	-0.36	NaN	0.43	0.34
Elapsed Time	0.04	-0.52	-0.36	1.00	NaN	-0.44	-0.52
Cycle	NaN	NaN	NaN	NaN	NaN	NaN	NaN
WhAccu	0.81	0.35	0.43	-0.44	NaN	1.00	0.37
Power	0.04	1.00	0.34	-0.52	NaN	0.37	1.00

Remaining Charging Time Features Correlation:

	Voltage	Current	Temperature	Elapsed Time	Cycle	Capacity	WhAccu	Power
Voltage	1.00	-0.29	0.07	0.06	-0.43	0.80	0.78	-0.22
Current	-0.29	1.00	0.27	-0.33	0.42	-0.28	-0.28	0.99
Temperature	0.07	0.27	1.00	-0.13	-0.21	0.24	0.24	0.27
Elapsed Time	0.06	-0.33	-0.13	1.00	-0.31	0.02	0.02	-0.33
Cycle	-0.43	0.42	-0.21	-0.31	1.00	-0.55	-0.55	0.40
Capacity	0.80	-0.28	0.24	0.02	-0.55	1.00	1.00	-0.23
WhAccu	0.78	-0.28	0.24	0.02	-0.55	1.00	1.00	-0.23
Power	-0.22	0.99	0.27	-0.33	0.40	-0.23	-0.23	1.00

Feature-to-feature correlations were examined to identify potential multicollinearity among modelling inputs.

Data Pipeline

- Data Quality Assessment — fits your missing-value/data-quality work.

- Feature Engineering — makes the pipeline clearer than only saying feature selection.

- File Organization — useful because you created organized outputs/reports during the project.

Raw Data
↓
Data Reading
↓
Cleaning & Preprocessing
↓
EDA & Data Understanding
↓
Feature Engineering / Feature Selection
↓
Target Definition
↓
Train–Validation–Test Preparation
↓
Correlation Analysis
↓
Multicollinearity Check
↓
Data Leakage Check
↓
Modeling-Ready Dataset

Conclusion

1. Dataset was cleaned, explored, and analysed through EDA.
2. Relevant features and Remaining Charging Time as the target were identified.
3. Train–test split and basic correlation, multicollinearity, and leakage checks were completed
4. Dataset is now ready for regression model development in the next stage.

Thank You!